Moment Generating Functions

1) Find the mean, variance and MGF of binomial distribution.

Sol:

P.M.F of binomial distribution:

$$P(X=x) = nC_x p^x q^{n-x} \qquad x = 0, 1, 2, \ldots n$$

where $p+q=1$

MGF is defined by

$$M_x(t) = \sum e^{tx} P(x) \Rightarrow E(e^{tx})$$

$$= \sum_{x=0}^{n} e^{tx} nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^{n} nC_x (pe^t)^x q^{n-x}$$

$$= nC_0 (pe^t)^0 q^n + nC_1 (pe^t)^1 q^{n-1}$$

$$+ \ldots + nC_r (pe^t)^r q^{n-r} + nC_n (pe^t)^n q^{n-n}$$

$$(a+b)^n = a^n + nC_1 a^{n-1} b + \ldots + nC_r a^r b^{n-r} + \ldots + b^n$$

$$M_x(t) = (q+pe^t)^n \quad //$$

$$\text{Mean} = M_x'(t)$$

$$M_x'(t) = \left[\frac{d}{dt}(q+pe^t)^n\right]_{t=0}$$

$$= n(q+pe^t)^{n-1} \times pe^t$$

$$= \left[np(q+pe^t)^{n-1} \times e^t\right]_{t=0}$$

$$= np(q+pe^0)^{n-1} \times e^0$$

$$= np(p+q)^{n-1}$$

$$\boxed{\text{Mean} = np}$$

$$M''_x(t) = \frac{d}{dt}\left[M_x'(t)\right]_{t=0}$$

$$= \frac{d}{dt}\left(npe^t(q+pe^t)^{n-1}\right)$$

$$= np\left[\frac{d}{dt}\, e^t(q+pe^t)^{n-1}\right]$$

$$= np\left(e^t \times (n-1)(q+pe^t)^{n-2} \times pe^t + (q+pe^t)^{n-1} e^t\right)_{t=0}$$

$$= np\left(e^0 \times (n-1)(q+p)^{n-2} \times p + (q+p)^{n-1}\right)$$

$$= np\left((n-1)p+1\right)$$

$$= np(np-p+1)$$

$$= n^2p^2 - np^2 + np$$

$$\text{Variance} = n^2p^2 - np^2 + np - (np)^2$$

$$= \cancel{n^2p^2} - np^2 + np - \cancel{n^2p^2}$$

$$= np - np^2$$

$$= np(1-p)$$

$$\boxed{\text{Variance} = npq}$$

Poisson distribution

Derive the MGF, mean and variance of Poisson distribution

Sol:

The p.mf of poisson distribution is

$$P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \qquad x = 0, 1, 2 \ldots\ldots \infty$$

M.G.F is defined by

$$M_x(t) = E(e^{tx})$$

$$= \sum_{x=0}^{\infty} e^{tx}\, p(x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(e^{t}\lambda)^{x}}{x!}$$

$$= e^{-\lambda}\left[1 + \frac{\lambda e^{t}}{1!} + \frac{(\lambda e^{t})^{2}}{2!} + \frac{(\lambda e^{t})^{3}}{3!} + \ldots\right]$$

$$e^{x} = \left[1 + \frac{x}{1!} + \frac{x^{2}}{2!} + \frac{x^{3}}{3!} + \ldots\right]$$

$$M_x(t) = e^{-\lambda} e^{\lambda e^{t}}$$

$$= e^{\lambda e^{t} - \lambda}$$

$$\boxed{M_x(t) = e^{\lambda(e^{t}-1)}}$$

$$M_x'(t) = \frac{d}{dt}\left[M_x(t)\right]_{t=0}$$

$$= \frac{d}{dt}\left(e^{\lambda(e^t-1)}\right)$$

$$= \left[e^{\lambda(e^t-1)} \times \lambda e^t\right]_{t=0}$$

$$= \left(e^{\lambda(1-1)} \times \lambda e^0\right)$$

$$= e^0 \times \lambda e^0$$

$$\boxed{M'_x(t) = \lambda}$$

$$\boxed{\text{Mean} = \lambda}$$

$$M''_x(t) = \frac{d}{dt}\left[M'_x(t)\right]_{t=0}$$

$$= \frac{d}{dt}\left[\lambda e^t e^{\lambda(e^t-1)}\right]$$

$$= \lambda\left[e^t\left[e^{\lambda(e^t-1)} \times \lambda e^t\right] + e^{\lambda(e^t-1)} e^t\right)_{t=0}$$

$$= \lambda\left(e^0\left(e^{\lambda(e^0-1)} \lambda e^0\right) + e^{\lambda(e^0-1)} e^0\right)$$

$$= \lambda\left(\lambda + 1\right)$$

$$M''_x(t) = \lambda^2 + \lambda$$

$$\text{Variance} = \lambda^2 + \lambda - (\lambda)^2$$

$$= \lambda^2 + \lambda - \lambda^2$$

$$\boxed{\text{Variance} = \lambda}$$

Geometric distribution

Derive MGF, mean, variance of Geometric distribution

Sol:

The density function of Geometric distribution

$$P(X=x) = q^{x-1} p \qquad x = 1, 2, \ldots \infty$$

where $p+q=1$

M.G.F

$$M_x(t) = E(e^{tx})$$

$$= \sum_{x=0}^{\infty} e^{tx} p(x)$$

$$= \sum_{x=0}^{\infty} e^{tx} q^{x-1} p$$

$$= \sum_{x=0}^{\infty} (e^t)^x (q^x) q^{-1} p$$

$$= \sum_{x=0}^{\infty} (qe^t)^x p/q$$

$$= p/q \left[(qe^t) + (qe^t)^2 + (qe^t)^3 + \ldots \right]$$

$$= \frac{p}{\cancel{q}} \cancel{q} e^t \left[1 + (qe^t) + (qe^t)^2 + \ldots \right]$$

$$= pe^t \left[1 + (qe^t) + (qe^t)^2 + \ldots \right]$$

$$\boxed{(1-x)^{-1} = 1 + x + x^2 + \ldots} = \frac{1}{1-x} = \frac{1}{1-qe^t}$$

$$M_x(t) = \frac{pe^t}{(1-qe^t)}$$

$$\boxed{M_x(t) = \frac{pe^t}{1-qe^t}}$$

$$M'x(t) = \frac{d}{dt}\left[\frac{pe^t}{1-qe^t}\right]_{t=0}$$

$$= \frac{(1-qe^t)\,pe^t - pe^t\,(0-qe^t)}{(1-qe^t)^2}$$

$$= \frac{pe^t - pqe^{2t} + pqe^{2t}}{(1-qe^t)^2}$$

$$M'x(t) = \frac{pe^t}{(1-qe^t)^2}$$

$$M''x(t) = \frac{(1-qe^t)^2\,pe^t - pe^t \cdot 2(1-qe^t)(0-qe^t)}{(1-qe^t)^4}$$

$$M'x(0) = \frac{p}{(1-q)^2} = \frac{p}{p^2} = 1/p$$

$$M''x(0) = \frac{(1-q)^2\,p - p \cdot 2(1-q)(-q)}{(1-q)^4}$$

$$= \frac{p^2 \cdot p + p \cdot 2 \times pq}{p^4}$$

$$= \frac{p^3}{p^4} + \frac{2p^2 q}{p^4}$$

$$M''(x) = \frac{1}{p} + \frac{2q}{p^2}$$

$$\text{Variance} = E(x^2) - (E(x))^2$$

$$= \frac{1}{p} + \frac{2q}{p^2} - 1/p^2$$

$$= \frac{q}{p^2}$$

Uniform distribution

Derive MGF mean and variance of uniform distribution

Sol:

Probability density function of r.v is

$$f(x) = \frac{1}{b-a} \quad , \quad a < x < b$$

M.G.F is defined by

$$M_x(t) = E(e^{tx})$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x)\, dx$$

$$= \int_{a}^{b} e^{tx} \frac{1}{b-a}\, dx$$

$$= \frac{1}{b-a} \int_{a}^{b} e^{tx}\, dx$$

$$= \frac{1}{b-a} \left[\frac{e^{tx}}{t}\right]_a^b$$

$$= \frac{1}{b-a} \left[\frac{e^{tb}}{t} - \frac{e^{ta}}{t}\right]$$

$$= \frac{1}{t} \frac{1}{b-a} \left(e^{bt} - e^{at}\right)$$

$$= \frac{1}{(b-a)t} \left[\left(1 + \frac{bt}{1!} + \frac{(bt)^2}{2!} + \cdots\right) - \left(1 + \frac{at}{1!} + \frac{(at)^2}{2!} + \cdots\right)\right]$$

$$= \frac{1}{(b-a)t} \left[\frac{(b-a)t}{1!} + \frac{(b^2-a^2)}{2!} t^2 + \cdots \frac{(b^3-a^3)t^3}{3!} + \cdots\right]$$

$$= \frac{1}{(b-a)t} \left[(b-a)t + \frac{(b-a)(b+a)}{2!} t^2 + \frac{(b-a)(b^2+ab+a^2)}{3!} + \cdots\right]$$

$$= \frac{(b-a)t}{(b-a)t} \left[1 + \frac{b+a}{2!} t + \frac{b^2+ab+a^2}{3!} t^2 + \cdots\right]$$

$$= 1 + \frac{(b+a)t}{2!} + \frac{(b^2+ab+b^2)t^2}{3!} + \cdots$$

$$\text{Mean} = E(x) = (\text{coeff. of } t) \text{ in } M_x(t) = \frac{b+a}{2!}$$

$$\boxed{E(x) = \frac{b+a}{2}}$$

$$E(x^2) = \left(\begin{array}{c}\text{coeff of } t^2 \\ \text{in } M_x(t)\end{array}\right) 2! = \frac{b^2+ab+a^2}{3!} 2! = \frac{b^2+ab+a^2}{3}$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$= \frac{b^2+ab+a^2}{3} - \frac{b^2+2ab+a^2}{4}$$

$$= \frac{4b^2+4ab+4a^2-3a^2-6ab-3a^2}{12}$$

$$= \frac{b^2-2ab+a^2}{12}$$

$$\text{Variance} = \frac{(b-a)^2}{12}$$

Normal distribution

Derive MGF, mean, variable of Normal distribution

Sol:

First, let us find MGF for standard normal distribution.

Density function of standard normal variable is

$$f(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2} \quad -\infty < z < \infty$$

where $\boxed{z = \frac{x-\mu}{\sigma}}$

MGF is defined as $M_Z(t) = E(e^{tz})$

$$= \int_{-\infty}^{\infty} e^{tz} f(z)\,dz$$

$$= \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{z^2}{2} - tz\right)}$$

The power $\left(\frac{z^2}{2} - tz\right)$ looks like $\boxed{A^2 - 2AB}$

So lets add and subtract $\frac{t^2}{2}$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{z^2}{2} - tz + \frac{t^2}{2} - t^2/2\right)}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left[\left(\frac{z}{\sqrt{2}} - t/\sqrt{2}\right)^2 - \frac{t^2}{2}\right]}\,dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{z-t}{\sqrt{2}}\right)^2} e^{t^2/2}\,dz$$

Let $u = \frac{z-t}{\sqrt{2}}$

$\frac{du}{dz} = \frac{1}{\sqrt{2}}$ $\quad dz = \sqrt{2}\,du$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2} e^{\frac{t^2}{2}} \sqrt{2}\,du$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2} e^{\frac{t^2}{2}}\,du$$

$$= \frac{e^{t^2/2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2}\,du$$

$$= \frac{e^{\frac{t^2}{2}}}{\cancel{\sqrt{\pi}}} \times \cancel{\sqrt{\pi}}$$

$$M_Z(t) = e^{t^2/2}$$

$$M_z(t) = E\left(e^{tz}\right) = e^{t^2/2}$$

$$E\left(e^{t\left(\frac{x-\mu}{\sigma}\right)}\right) = e^{t^2/2}$$

$$E\left(e^{\frac{tx}{\sigma}}\, e^{-t\mu/\sigma}\right) = e^{t^2/2}$$

x is the random variable

$$E\left(e^{x\, t/\sigma}\right) = e^{t^2/2}\, e^{t\mu/\sigma}$$

Let $t = t/\sigma$

$$E\left(e^{xt}\right) = e^{\frac{t^2\sigma^2}{2}}\, e^{t\mu}$$

$$\therefore M_x(t) = E\left[e^{xt}\right] = e^{\frac{\sigma^2 t^2}{2}}\, e^{t\mu}$$

$$= \left[1 + \frac{\frac{\sigma^2 t^2}{2}}{1!} + \frac{\frac{\sigma^4 t^4}{4}}{2!} + \cdots\right]\left[1 + \frac{t\mu}{1!} + \frac{t^2\mu^2}{2!} + \cdots\right]$$

$$= 1 + \frac{t\mu}{1!} + \frac{t^2\mu^2}{2!} + \frac{\sigma^2 t^2}{2} + \cdots$$

E(x) Mean = coeff of $t = \mu$

$$E(x^2) = (\text{coeff of } t^2)\,2! = \left(\frac{\mu^2 + \sigma^2}{2}\right) 2!$$

$$= \mu^2 + \sigma^2$$

$$\text{Variance} = \mu^2 + \sigma^2 - (\mu)^2$$

$$\boxed{\text{Variance} = \sigma^2}$$

Exponential distribution

Derive MGF, variance, mean of exponential distribution

The density function of exponential distribution

$$f(x) = \lambda e^{-\lambda x} \quad x > 0$$

MGF:

$$M_x(t) = \int_0^\infty \lambda e^{-\lambda x} e^{xt} \, dx$$

$$= \lambda \int_0^\infty e^{-\lambda x + xt} \, dx$$

$$= \lambda \int_0^\infty -e^{x(-t+\lambda)} \, dx$$

$$\cancel{= \lambda \left[\frac{e^{(t-\lambda)x}}{t-\lambda} \right]_0^\infty}$$

$$= \cancel{\lambda} \left[\frac{e^{-x(\lambda - t)}}{-(\lambda - t)} \right]_0^\infty$$

$$= \lambda \left[0 - \frac{1}{-(\lambda - t)} \right]$$

$$= \lambda \left[\frac{1}{\lambda - t} \right]$$

$$\boxed{M_x(t) = \frac{\lambda}{\lambda - t}}$$

To derive the mean and variance expand $M_x(t)$ in power of t

$$M_x(t) = \frac{\cancel{\lambda}}{\cancel{\lambda}(1 - t/\lambda)}$$

$$= \frac{1}{1 - t/\lambda}$$

$$= (1 - t/\lambda)^{-1}$$

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots$$

$$\left(1 - t/\lambda\right)^{-1} = 1 + \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \frac{t^3}{\lambda^3} + \cdots$$

Mean = coefficient of t

$$E(x) = \frac{1}{\lambda}$$

$E(x^2)$ ~~Variance~~ = coeff of $t^2 \times 2!$

$$= 1/\lambda^2 \times 2$$

$$E(x^2) = 2/\lambda^2$$

$$E(x^2) - (E(x))^2 = \frac{2}{\lambda^2} - 1/\lambda^2$$

$$\boxed{\text{Variance} = 1/\lambda^2}$$

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MGF : $M_x(t) = E(e^{tx})$ (or) $\sum_{r=0}^{\infty} \frac{t^r}{r!} \mu'_r$

Discrete Distribution:

a) Binomial : $p(x) = nC_x \, p^x q^{n-x}$

$\mu = np$

$\sigma^2 = npq$

$M_x(t) = (q+pe^t)^n$

b) Poisson λ : $p(x) = e^{-\lambda} \lambda^x / x!$

$\mu = \sigma^2 = \lambda$

$M_x(t) = e^{\lambda(e^t - 1)}$

c) Geometric : $p(x) = q^x p$ Memoryless

$\mu = q/p$

$\sigma^2 = q/p^2$

$M_x(t) = p / 1-qe^t$

Continuous Distribution:

a) Uniform / Rectangular : $f(x) = \frac{1}{b-a}$, $a \le x < b$

$\mu = a+b/2$

$\sigma^2 = (b-a)^2/12$

$M_x(t) = \frac{e^{bt} - e^{at}}{t(b-a)}$

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Memorly

b) Exponential : $f(x) = \lambda e^{-\lambda x}$ $\quad x \geq 0$

$\mu' = 1/\lambda$

$\mu'_r = r!/\lambda^r$

$\sigma^2 = 1/\lambda^2$

$M_x(t) = \lambda/\lambda - t \quad t < \lambda$

c) Normal : $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$-\infty < x, \mu < \infty$
$\sigma > 0$

$Z = \frac{X-\mu}{\sigma}$

Standard Deviation $= \sqrt{\sigma^2} = \sigma$

$M_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

Functions of R.V :

$$f_y(y) = f_x(x) \left| \frac{dx}{dy} \right|$$